

SCQ (Single Correct Type) :

- The number of value(s) of x satisfying the equation $(2011)^x + (2012)^x + (2013)^x - (2014)^x = 0$ is/are :
 (A) Exactly 2 (B) Exactly 1 (C) More than one (D) 0
- Let $A = [a, \infty)$ denotes domain, then $f : [a, \infty) \rightarrow B$, $f(x) = 2x^3 - 3x^2 + 6$ will have an inverse for the smallest real value of a , if:
 (A) $a = 1, B = [5, \infty)$ (B) $a = 2, B = [10, \infty)$ (C) $a = 0, B = [6, \infty)$ (D) $a = -1, B = [1, \infty)$
- Which of the following is not an odd function ?
 (A) $\ln\left(\frac{x^4 + x^2 + 1}{(x^2 + x + 1)^2}\right)$
 (B) $\text{sgn}(\text{sgn}(x))$
 (C) $\sin(\tan x)$
 (D) $f(x)$, where $f(x) + f\left(\frac{1}{x}\right) = f(x) \cdot f\left(\frac{1}{x}\right) \quad \forall x \in \mathbb{R} - \{0\}$ and $f(2) = 33$
- The function, $f(x) = \begin{cases} \frac{(x^{2n})}{(x^{2n} \text{sgn } x)^{2n+1}} \left(\frac{e^x - e^{-x}}{e^x + e^{-x}} \right) & x \neq 0 \\ 1 & x = 0 \end{cases} \quad n \in \mathbb{N}$ is :
 (A) Odd function (B) Even function
 (C) Neither odd nor even function (D) Constant function
- Consider all function $f : \{1, 2, 3, 4\} \rightarrow \{1, 2, 3, 4\}$ which are one-one, onto and satisfy the following property :
 if $f(k)$ is odd then $f(k+1)$ is even, $k = 1, 2, 3$.
 The number of such function is :
 (A) 4 (B) 8 (C) 12 (D) 16
- Find the sum of the infinite series $\frac{1}{9} + \frac{1}{18} + \frac{1}{30} + \frac{1}{45} + \frac{1}{63} + \dots$
 (A) $\frac{1}{3}$ (B) $\frac{1}{4}$ (C) $\frac{1}{5}$ (D) $\frac{2}{3}$

MCQ (One or more than one correct) :

7. If the equation $\cot^4 x - 2 \operatorname{cosec}^2 x + a^2 = 0$ has at least one solution then possible integral values of a can be :
 (A) -1 (B) 0 (C) 1 (D) 2
8. Let $k = \frac{(2\tan 4 + 3\cot 4)(2\cot 4 + 3\tan 4)}{24\cot^2 8 + 25}$ and $f(x) = 8x^2 + (\alpha + 2)x - 3\alpha$. If k lies between the roots of the equation $f(x) = 0$ then
 (A) $k = 1$ (B) $k = 2$
 (C) the least integral value of α is 6. (D) The least integral value of α is 37.
9. The possible positive integral value of 'k' for which $5x^2 - 2kx + 1 < 0$ has exactly one integral solution may be divisible by :
 (A) 2 (B) 3 (C) 5 (D) 7
10. If a, b are two numbers such that $a^2 + b^2 = 7$ and $a^3 + b^3 = 10$, then:
 (A) The greatest value of $|a + b| - 5$ (B) The greatest value of $(a + b)$ is 4
 (C) The least value of $(a + b)$ is 1 (D) The least value of $|a + b|$ is 1
11. If $\left(\frac{a^3}{a-1}, \frac{a^2-3}{a-1}\right); \left(\frac{b^3}{b-1}, \frac{b^2-3}{b-1}\right); \left(\frac{c^3}{c-1}, \frac{c^2-3}{c-1}\right)$ lie on $L : lx + my + n = 0$; where a, b, c are real numbers different from 1; then
 (A) $a + b + c = \frac{-m}{l}$ (B) $abc = \frac{m+n}{l}$
 (C) $ab + bc + ca = \frac{n}{l}$ (D) $abc - (ab + bc + ca) + 3(a + b + c) = 0$
12. If $f(x) = 2x + |x|$, $g(x) = \frac{1}{3}(2x - |x|)$ and $h(x) = f(g(x))$, then $\underbrace{h(h(h(\dots(h(x))))}_{h \text{ repeated } n \text{ times}})$ is
 (A) identity function (B) one-one (C) odd (D) periodic
13. Let $f(x) = \begin{cases} x^2 & 0 < x < 2 \\ 2x - 3 & 2 \leq x < 3 \\ x + 2 & x \geq 3 \end{cases}$, then the true equations :
 (A) $f\left(f\left(f\left(\frac{3}{2}\right)\right)\right) = f\left(\frac{3}{2}\right)$ (B) $1 + f\left(f\left(f\left(\frac{5}{2}\right)\right)\right) = f\left(\frac{5}{2}\right)$
 (C) $f(f(f(2))) = f(1)$ (D) $\underbrace{f(f(f(\dots f(4)\dots))}_{1004 \text{ times}}) = 2012$
14. Let $f_n(x) = (\sin x)^{1/n} + (\cos x)^{1/n}$, $x \in \mathbb{R}$, then :
 (A) $f_2(x) > 1$ for all $x \in \left(2k\pi, (4k+1)\frac{\pi}{2}\right)$, $k \in \mathbb{I}$
 (B) $f_2(x) = 1$ for $x = 2k\pi$, $k \in \mathbb{I}$
 (C) $f_2(x) > f_3(x)$ for all $x \in \left(2k\pi, (4k+1)\frac{\pi}{2}\right)$, $k \in \mathbb{I}$

(D) $f_3(x) \geq f_5(x)$ for all $x \in \left(2k\pi, (4k+1)\frac{\pi}{2}\right), k \in I$

(Where I denotes set of integers)

15. Consider the sequence of numbers $\alpha_0, \alpha_1, \dots, \alpha_n$ where $\alpha_0 = 17.23$, $\alpha_1 = 33.23$ and $\alpha_{r+2} = \frac{\alpha_r + \alpha_{r+1}}{2}$.

Then

(A) $|\alpha_{10} - \alpha_9| = \frac{1}{32}$

(B) $\alpha_0 - \alpha_1, \alpha_1 - \alpha_2, \alpha_2 - \alpha_3, \dots$ are in G.P.

(C) $\alpha_0 - \alpha_2, 2(\alpha_1 - \alpha_2), \alpha_1 - \alpha_3$ are in H.P.

(D) $|\alpha_{10} - \alpha_9| = |\alpha_8 - \alpha_7|$

Numerical Based Questions :

16. Given that for $a, b, c, d \in \mathbb{R}$, if $a \sec(200^\circ) - c \tan(200^\circ) = d$ and $b \sec(200^\circ) + d \tan(200^\circ) = c$, then

find the value of $\left(\frac{a^2 + b^2 + c^2 + d^2}{bd - ac}\right) \sin 20^\circ$.

17. The numbers $\frac{1}{3}, \frac{1}{3} \log_x y, \frac{1}{3} \log_y z, \frac{1}{7} \log_z x$ are in H.P. If $y = x^r$ and $z = x^s$, then $4(r + s) =$

18. The ordered pair (x, y) satisfying the equation

$$x^2 = 1 + 6 \log_4 y \text{ and } y^2 = 2^x y + 2^{2x+1}$$

are (x_1, y_1) and (x_2, y_2) , then find the value of $\log_2 |x_1 x_2 y_1 y_2|$.

19. Let $p(x) = x^2 - x + 1$. Let α be a root of $p(p(p(p(x))))$. Find the value of $|(p(\alpha) - 1)p(\alpha)p(p(\alpha))p(p(p(\alpha)))|$

20. The number of roots of equation:

$$\left(\frac{(x-1)(x-3)}{(x-2)(x-4)} - e^x\right) \left(\frac{(x+1)(x+3)e^x}{(x+2)(x+4)} - 1\right) (x^3 - \cos x) = 0$$

21. If a, b, c and d are positive integers and $a < b < c < d$ such that a, b, c are in A.P. and b, c, d are in G.P.

and $d - a = 30$. Find the value of sum of digits of $\frac{a+b+c+d}{3}$.

Paragraph Type Question:

Paragraph for Question Nos. 22 to 23

There are two sets A and B each of which consists of three numbers in A.P. whose sum is 15. D and d are their respective common differences such that $D - d = 1$, $D > 0$. If $\frac{p}{q} = \frac{7}{8}$ where p and q are the

product of the numbers in those sets A and B respectively

22. Sum of the product of the numbers in set A taken two at a time is :

(A) 51

(B) 71

(C) 74

(D) 86

23. Sum of the product of the numbers in set B taken two at a time is :

(A) 52

(B) 54

(C) 64

(D) 74

Match the column :

24.

	Column-I		Column-II
(A)	Minimum value of $y = 4 \sec^2 x + \cos^2 x$ for permissible real values of x is equal to	(P)	2
(B)	If m, n are positive integers and $m + n\sqrt{2} = \sqrt{41 + 24\sqrt{2}}$ then $(m + n)$ is equal to :	(Q)	7
(C)	Number of solutions of the equation : $\log_{\left(\frac{9x-x^2-14}{7}\right)}(\sin 3x - \sin x) = \log_{\left(\frac{9x-x^2-14}{7}\right)} \cos 2x$ is equal to	(R)	4
(D)	Consider an arithmetic sequence of positive integers. If the sum of the first ten terms is equal to the 58th term, then the least possible value of the first term is equal to :	(S)	5
		(T)	3

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ANSWER KEY OF CAPS 1+

1.	(B)	2.	(A)	3.	(D)	4.	(B)	5.	(C)	6.	(A)	7.	(ABC)
8.	(AC)	9.	(AC)	10.	(ABD)	11.	(ACD)	12.	(ABC)	13.	(ABCD)	14.	(AB)
15.	(ABC)	16.	2	17.	6	18.	7	19.	1	20.	7	21.	7
22.	(B)	23.	(D)	24.	[A → S; B → Q; C → P; D → R]								